

Deploying the portal to Vercel

A free way to put the site on the internet so other people can look at it.

What can and cannot go on Vercel

Vercel runs **serverless functions**: short-lived, no permanent disk. That suits the website and rules out the collector.

Piece	Vercel?	Why
Web portal, JSON API	Yes	Read-only, touches no files
PostgreSQL	No	Vercel has no database; use Neon
Collector and scheduler	No	One crawl runs over an hour; functions get seconds
Raw document archive	No	No persistent filesystem

So the shape is:

```
graph LR
    V[Vercel  
the website] -- reads --> N[Neon (PostgreSQL)]
    C[collector  
(your laptop,  
or a VPS later)] -- writes --> N
```

The collector keeps running wherever you run it and writes to the same hosted database the website reads from. Your laptop being off stops new listings arriving; it does not take the site down.

1. Create the database (Neon, free)

1. Sign up at neon.tech and create a project. Choose the **Singapore** or **Mumbai** region — closest to both your users and Vercel's Asian edge.
2. Copy the connection string. Take the **pooled** one, whose host contains `-pooler`. It looks like:

```
postgresql://user:pass@ep-xxx-pooler.ap-southeast-1.aws.neon.tech/neondb?sslmode=require
```

3. Change the scheme so SQLAlchemy uses psycopg 3:

```
postgresql+psycopg://user:pass@ep-xxx-pooler.../neondb?sslmode=require
```

Keep that string somewhere safe. It is a password.

2. Create the schema and load data

From your machine, pointing at Neon rather than the local database:

```
$env:DATABASE_URL = "postgresql+psycopg://...your neon url..."

.\.venv\Scripts\python.exe -m alembic upgrade head
.\.venv\Scripts\python.exe -m auction_portal crawl baanknet --limit 500
```

That creates the tables and puts real listings in them. The crawl takes about 25 minutes at 500 URLs.

Setting `DATABASE_URL` this way only affects that one PowerShell window. Close it and you are back on the local database.

3. Push to GitHub

If the repository has no remote yet:

```
git remote add origin https://github.com/YOUR-USERNAME/YOUR-REPO.git
git branch -M main
git push -u origin main
```

Nothing sensitive is committed — `.env`, `data/`, `.pgsql/` and `logs/` are all ignored. Check with `git status` before pushing if you want to be sure.

4. Connect Vercel

1. Go to vercel.com, sign in with GitHub.
2. **Add New** → **Project**, and pick the repository.
3. Leave the framework preset as **Other**. `vercel.json` already tells Vercel what to do.
4. Before deploying, open **Environment Variables** and add:

Name	Value
<code>DATABASE_URL</code>	your Neon pooled connection string
<code>CONTACT_EMAIL</code>	a real address you monitor
<code>CONTACT_URL</code>	<code>https://your-project.vercel.app/bot</code>
<code>DB_SERVERLESS</code>	<code>true</code>
<code>ENVIRONMENT</code>	<code>production</code>

`DB_SERVERLESS=true` matters. Without it each function instance holds a connection pool and Neon's connection limit is reached quickly.

5. **Deploy**. The first build takes two or three minutes.

5. Check it

Visit your `https://your-project.vercel.app`. Then:

```
.\.venv\Scripts\python.exe scripts\smoke_check.py https://your-project.vercel.app
.\.venv\Scripts\python.exe scripts\verify_manual.py https://your-project.vercel.app
```

Both run against the deployed site, so they check the real thing rather than your laptop.

Keeping it updated

The website is now public but **static in content** — it shows whatever is in Neon. To add listings, run the collector against Neon:

```
$env:DATABASE_URL = "postgresql+psycpg://...your neon url..."
.\.venv\Scripts\python.exe -m auction_portal schedule
```

Leave that running and the live site keeps gaining listings. Close it and the site stays up, simply stops growing.

Pushing to GitHub redeploys the site automatically. The database is untouched by deployments.

When to stop using this arrangement

It is a good way to show people the portal. It is not the final shape, for one reason: **the collector depends on someone's laptop**. That is fine for testing and wrong for a service people pay for.

The fix is a small VPS, around ₹800–2,000 a month, running `docker-compose.yml` — website, database and collector together, always on. See [DEPLOYMENT.md](#). At that point Vercel and Neon can either stay as the public face or be dropped entirely.

Things that will bite you

Cold starts. An idle Vercel function sleeps. The first visit after a quiet spell takes a few seconds. Normal, and free tiers cannot avoid it.

Free tier limits. Neon's free tier suspends the database after inactivity and has a storage cap. Enough for testing, not for a launch.

The 90-day retention job will not run, because the scheduler is not on Vercel. Run `maintain` occasionally against Neon, or move the collector to a server.

Do not put the legal review off. A site on the public internet with a real domain is materially different from a laptop nobody can reach. See Part E of the main document.